



懂设计懂工艺

贵州理科微波科技有限公司 电桥产品手册

贵州理科微波科技有限公司

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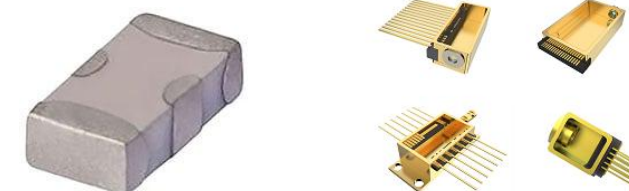
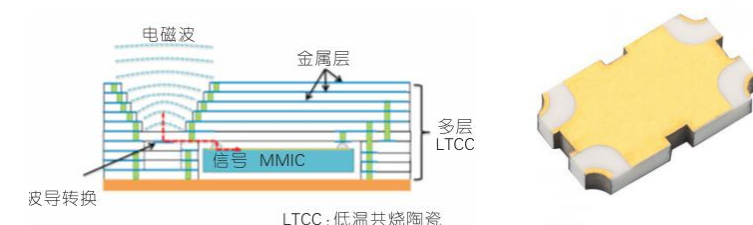
地址：贵州省遵义市汇川区上海路街道上海路新长征尊城
8号楼1203厂房

产品

- ❖ 滤波器
- ❖ 天线
- ❖ 电桥
- ❖ 电桥
- ❖ 封装

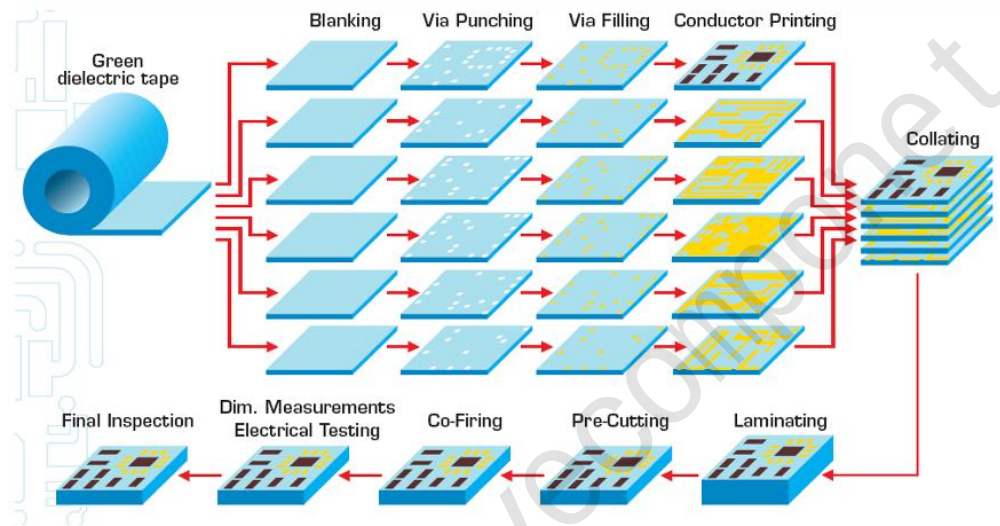
工艺

- ❖ PCB
- ❖ LTCC
- ❖ GaAs
- ❖ 薄膜
- ❖ 厚膜



电桥选型指南

提供基于LTCC（低温共烧陶瓷）和PCB工艺的电桥系列产品，在性能与可靠性上直接对标Mini-Circuits等国际领先品牌。致力于为高频、高速应用提供小型化、高抑制、低插损的优质选择。



电桥

序号	型号	频率/MHz	尺寸/mm	工艺	对标Mini型号
1	GLHC-100-160-32-25	100-160	3.2×2.5	LTCC	QCV-151
2	GLHC-270-520-32-16	270-520	3.2×1.6	LTCC	
3	GLHC-480-780-32-16	480-780	3.2×1.6	LTCC	
4	GLHC-780-1500-32-16	780-1500	3.2×1.6	LTCC	QCN-12
5	GLHC-1100-1900-32-16	1100-1900	3.2×1.6	LTCC	QCN-19
6	GPHC-1700-2300-20-12	1700-2300	2.0×1.25	PCB	QCN-25
7	GLHC-1800-2400-32-25	1800-2400	3.2×1.6	LTCC	
8	GPHC-2000-2600-20-12	2000-2600	2.0×1.25	PCB	
9	GPHC-3000-4000-20-12	3000-4000	2.0×1.25	PCB	
10	GPHC-3500-6000-20-12	3500-6000	2.0×1.25	PCB	

命名规则：以GLHC-100-160-32-25为例。

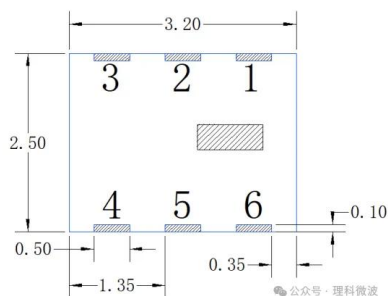
序号	类别	含义
1	G	公司代号
2	L/P	L代表LTCC工艺；P代表PCB工艺
3	HC	电桥
4	100-160	通带频率95MHz-160MHz
5	32-25	3.2mm×2.5mm

GLHC-100-160-32-25

LTCC 3dB 90° Hybrid Coupler

1.Features

- 1210 (3.2×2.5mm) Stand Package
- Continuous wave 2W
- Operating temperature: -55℃ to +125℃

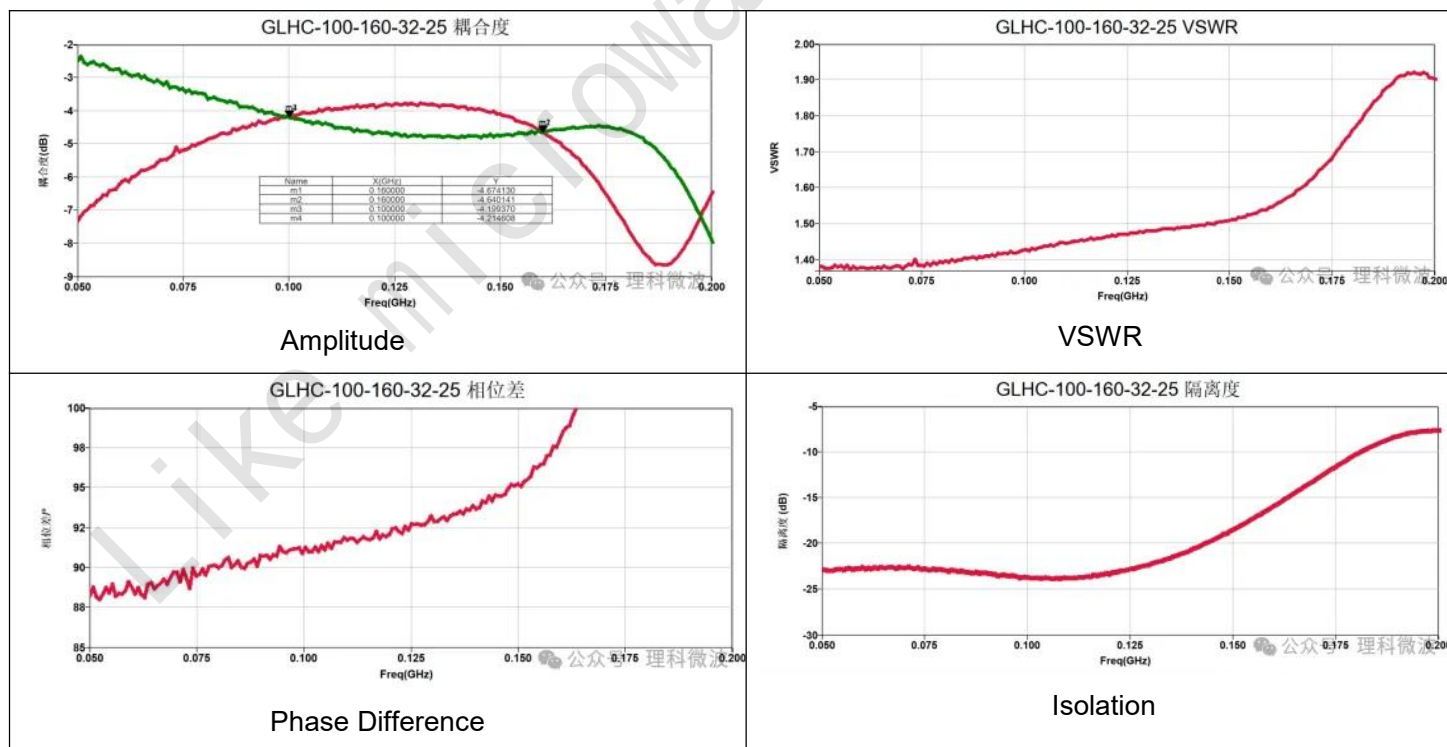


Function	Terminal
Sum Port	1
Port 1(0°)	4
Port 2(90°)	6
Isolation	3
GND	2,5

2.Typical Specifications

Parameter	Frequency (MHz)	SPEC		
		Min.	TYP	Max.
Insertion Loss (dB)	100~160		0.6	0.9
VSWR			1.4	1.8
Amplitude Unbalance(dB)			0.6	1.0
Phase Unbalance (Degrees)			8	11
Isolation			17	14

3.Electrical Performance Specifications

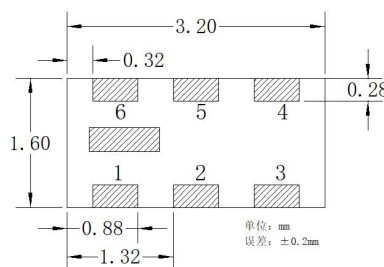


GLHC-270-520-32-16

LTCC 3dB 90° Hybrid Coupler

1.Features

- 1206 (3.2×1.6mm) Stand Package
- Continuous wave 2W
- Operating temperature: -55℃ to +125℃

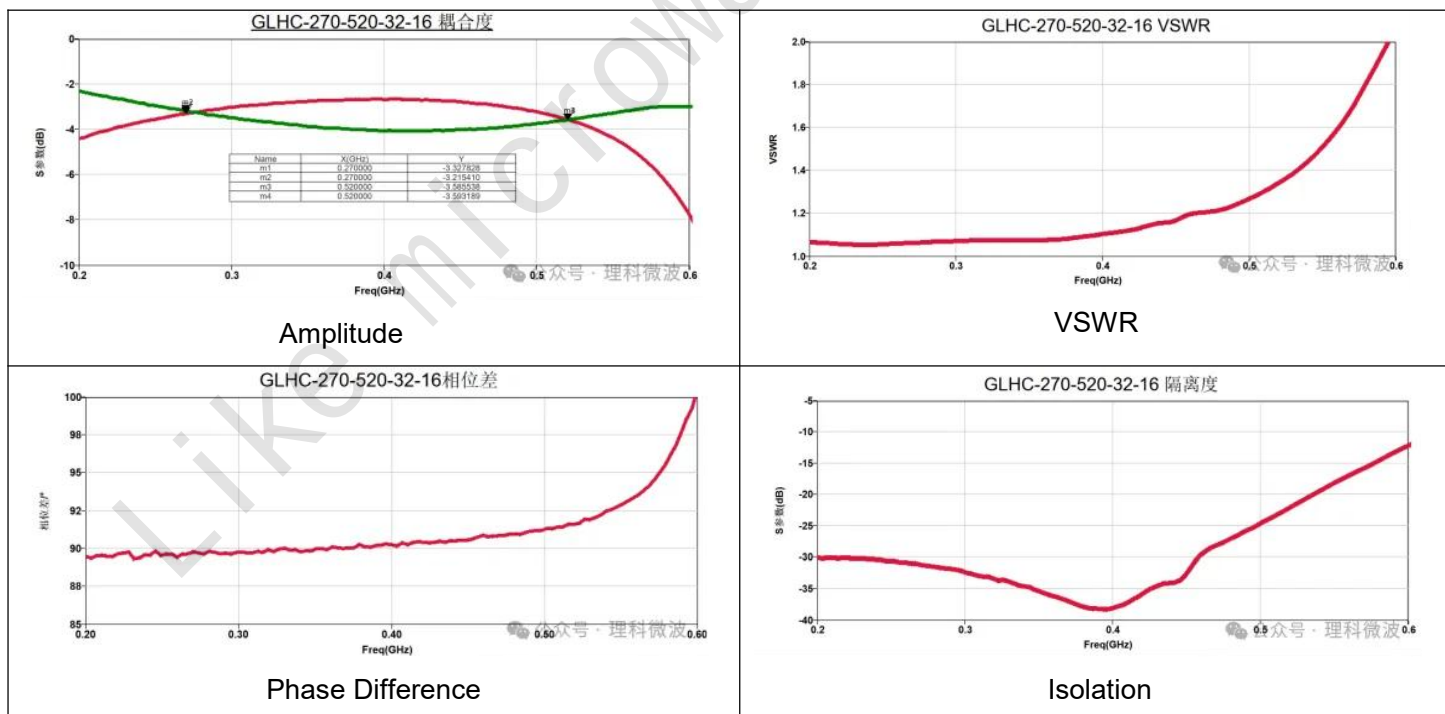


Function	Terminal
Sum Port	1
Port 1(0°)	4
Port 2(90°)	6
Isolation	3
GND	2,5

2.Typical Specifications

Parameter	Frequency (MHz)	SPEC		
		Min.	TYP	Max.
Insertion Loss (dB)	270~520		0.6	0.9
VSWR			1.4	1.8
Amplitude Unbalance(dB)			1.4	2.0
Phase Unbalance (Degrees)			3	5
Isolation			20	17

3.Electrical Performance Specifications

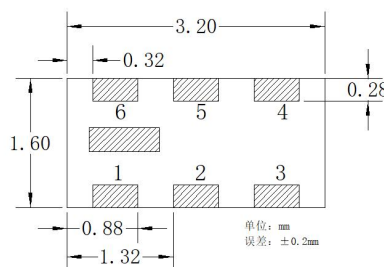


GLHC-480-780-32-16

LTCC 3dB 90° Hybrid Coupler

1.Features

- 1206 (3.2×1.6mm) Stand Package
- Continuous wave 2W
- Operating temperature: -55℃ to +125℃

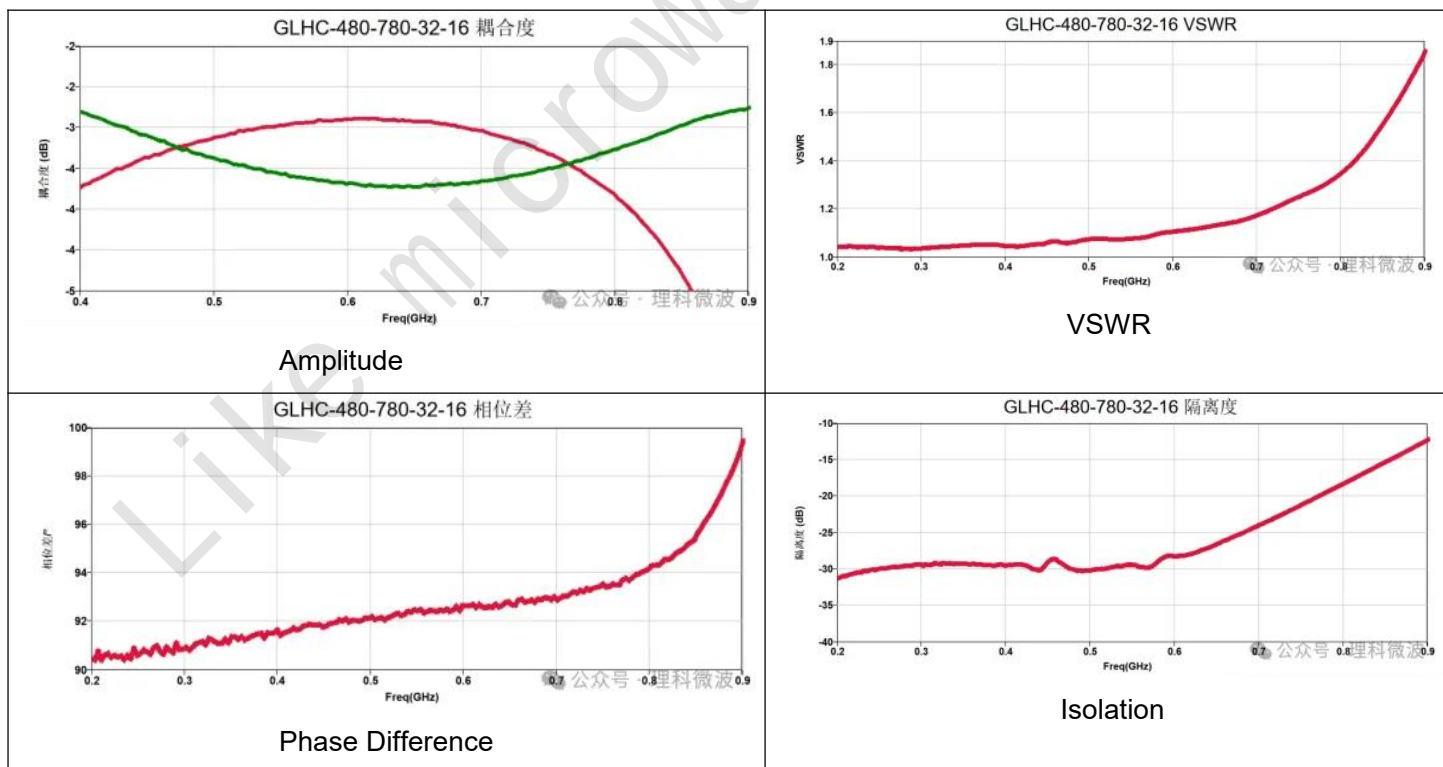


Function	Terminal
Sum Port	1
Port 1(0°)	4
Port 2(90°)	6
Isolation	3
GND	2,5

2.Typical Specifications

Parameter	Frequency (MHz)	SPEC		
		Min.	TYP	Max.
Insertion Loss (dB)	480~780		0.6	0.9
VSWR			1.4	1.8
Amplitude Unbalance(dB)			0.8	1.2
Phase Unbalance (Degrees)			3	5
Isolation			20	15

3.Electrical Performance Specifications

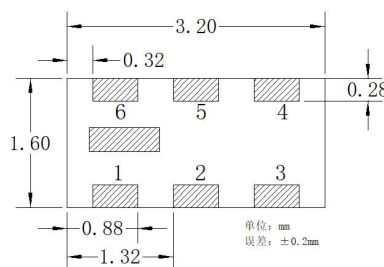


GLHC-780-1500-32-16

LTCC 3dB 90° Hybrid Coupler

1.Features

- 1206 (3.2×1.6mm) Stand Package
- Continuous wave 2W
- Operating temperature: -55℃ to +125℃

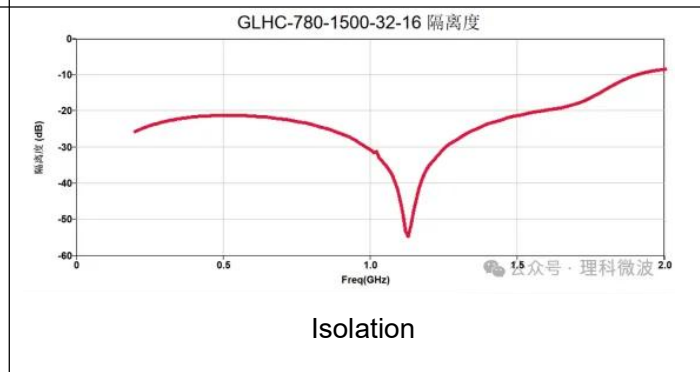
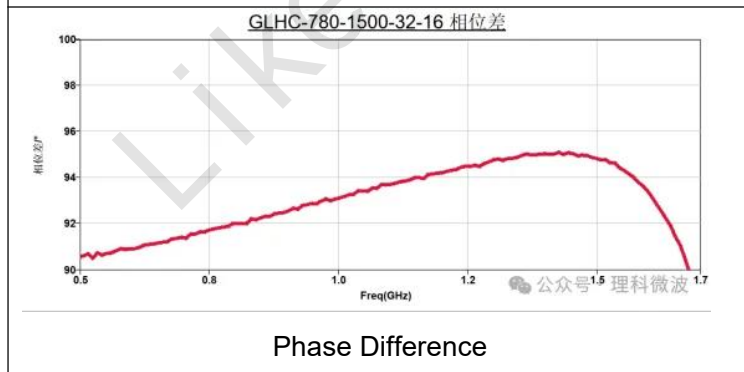
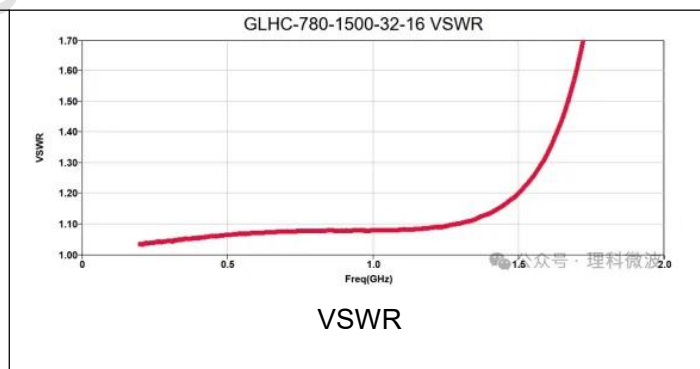
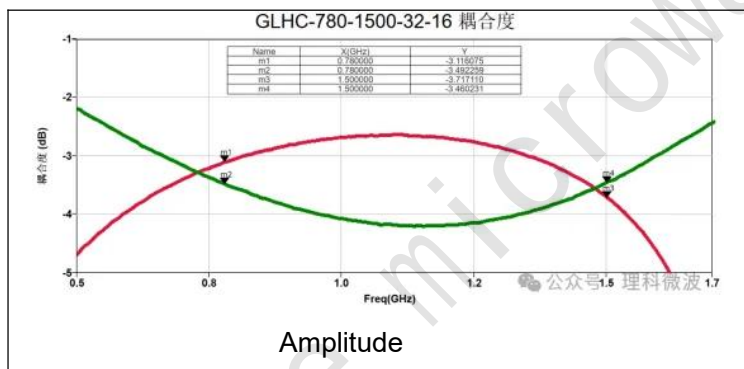


Function	Terminal
Sum Port	1
Port 1(0°)	4
Port 2(90°)	6
Isolation	3
GND	2,5

2.Typical Specifications

Parameter	Frequency (MHz)	SPEC		
		Min.	TYP	Max.
Insertion Loss (dB)	780~1500		0.6	0.9
VSWR			1.4	1.8
Amplitude Unbalance(dB)			0.8	1.2
Phase Unbalance (Degrees)			3	5
Isolation			20	15

3.Electrical Performance Specifications

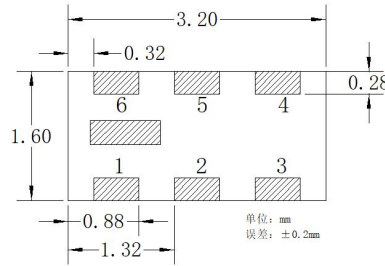


GLHC-1100-1900-32-16

LTCC 3dB 90° Hybrid Coupler

1.Features

- 1206 (3.2×1.6mm) Stand Package
- Continuous wave 2W
- Operating temperature: -55℃ to +125℃

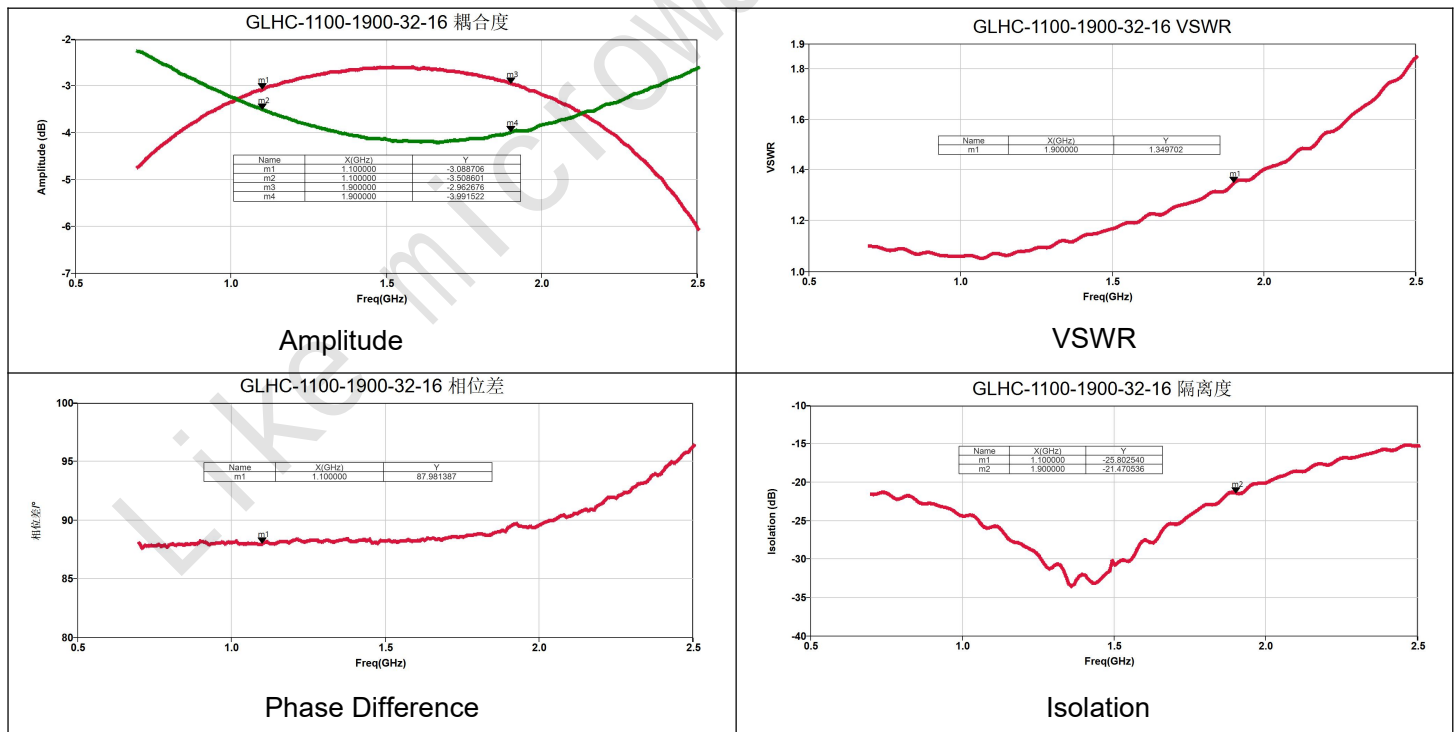


Function	Terminal
Sum Port	1
Port 1(0°)	4
Port 2(90°)	6
Isolation	3
GND	2,5

2.Typical Specifications

Parameter	Frequency (MHz)	SPEC		
		Min.	TYP	Max.
Insertion Loss (dB)	1100~1900		0.6	0.9
VSWR			1.4	1.8
Amplitude Unbalance(dB)			0.8	1.2
Phase Unbalance (Degrees)			3	5
Isolation			20	15

3.Electrical Performance Specifications



GPHC-1700-2300-20-12

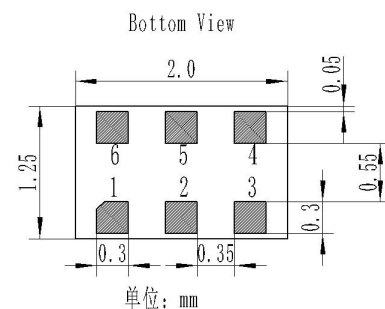
PCB 3dB 90° Hybrid Coupler

1.Features

- 0805 (2.0×1.25mm) Stand Package
- Continuous wave 2W
- Operating temperature: -55℃ to +125℃

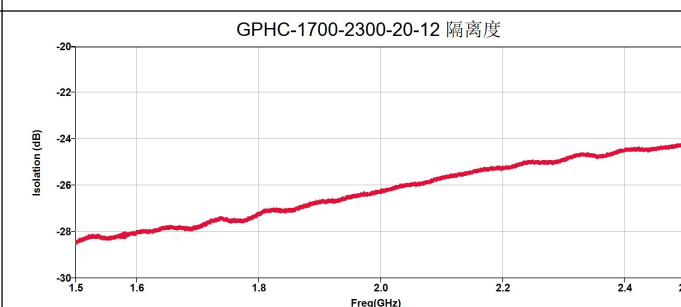
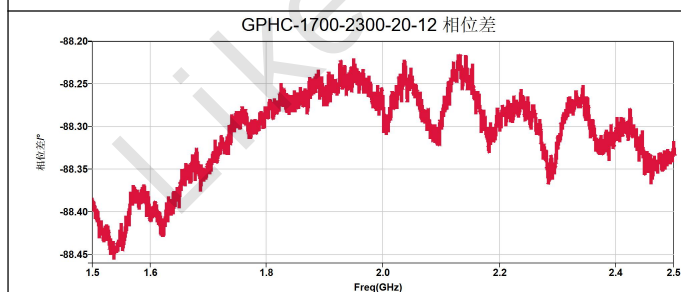
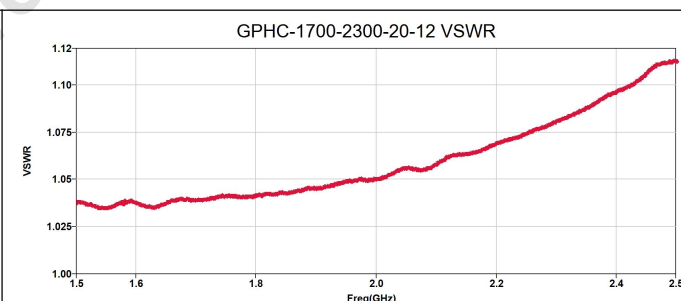
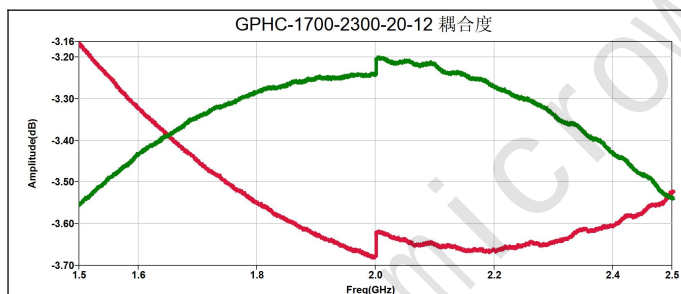
2.Typical Specifications

Function	Terminal	Function	Terminal
Sum Port	1	Isolation	3
Port 1(0°)	4	Port 2(90°)	6



Parameter	Frequency (MHz)	SPEC		
		Min.	TYP	Max.
Insertion Loss (dB)	1700~2300		0.6	0.9
VSWR			1.4	1.8
Amplitude Unbalance(dB)			0.8	1.2
Phase Unbalance (Degrees)			3	5
Isolation			20	15

3.Electrical Performance Specifications

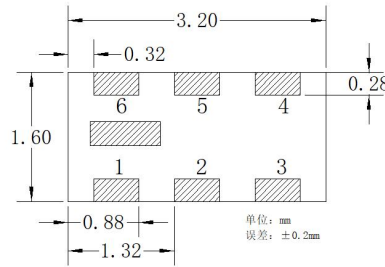


GLHC-1800-2400-32-16

LTCC 3dB 90° Hybrid Coupler

1.Features

- 1206 (3.2×1.6mm) Stand Package
- Continuous wave 2W
- Operating temperature: -55℃ to +125℃

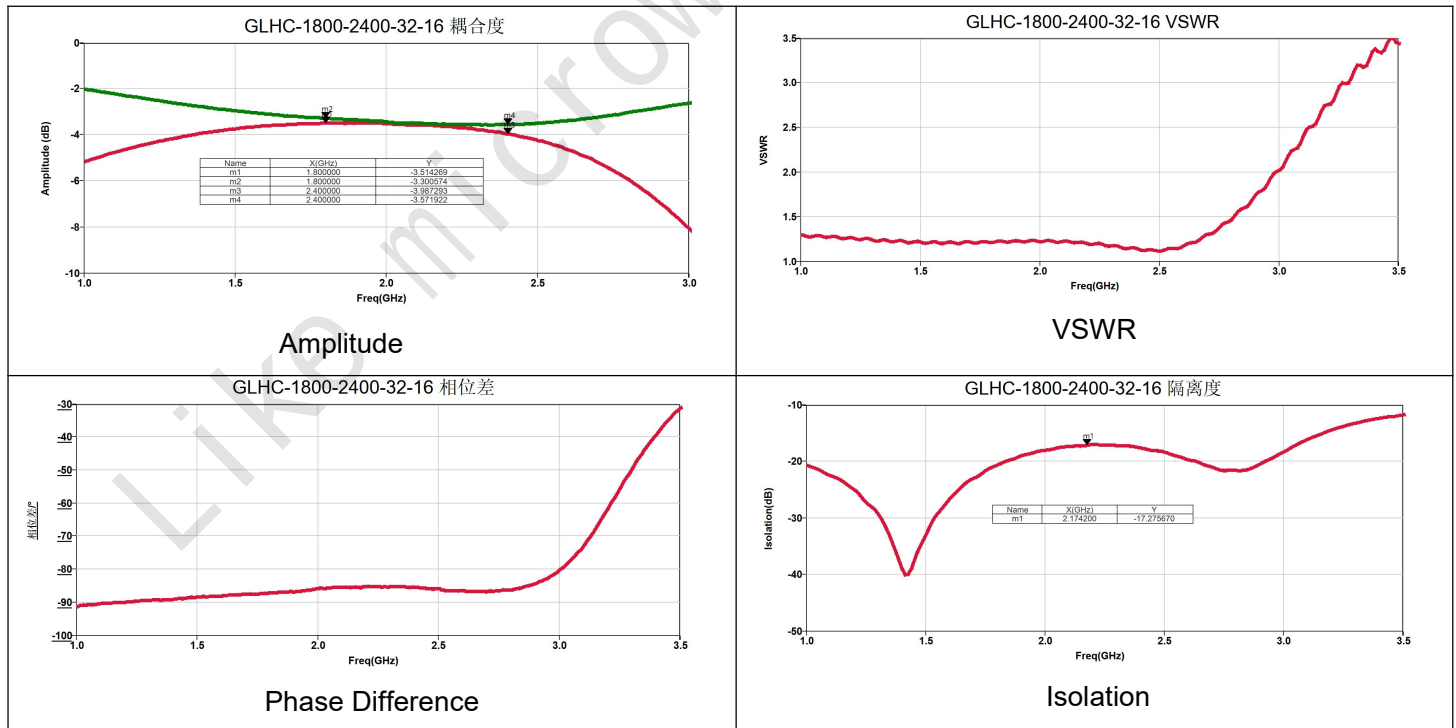


Function	Terminal
Sum Port	1
Port 1(0°)	4
Port 2(90°)	6
Isolation	3
GND	2,5

2.Typical Specifications

Parameter	Frequency (MHz)	SPEC		
		Min.	TYP	Max.
Insertion Loss (dB)	1800~2400		0.6	0.9
VSWR			1.4	1.8
Amplitude Unbalance(dB)			0.8	1.2
Phase Unbalance (Degrees)			3	5
Isolation			20	15

3.Electrical Performance Specifications

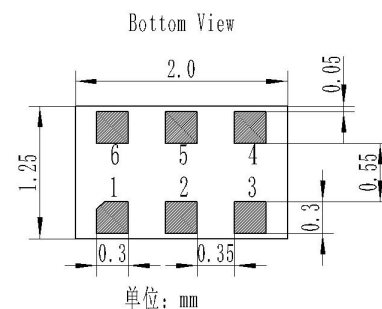


GPHC-2000-2600-20-12

PCB 3dB 90° Hybrid Coupler

1.Features

- 0805 (2.0×1.25mm) Stand Package
- Continuous wave 2W
- Operating temperature: -55℃ to +125℃

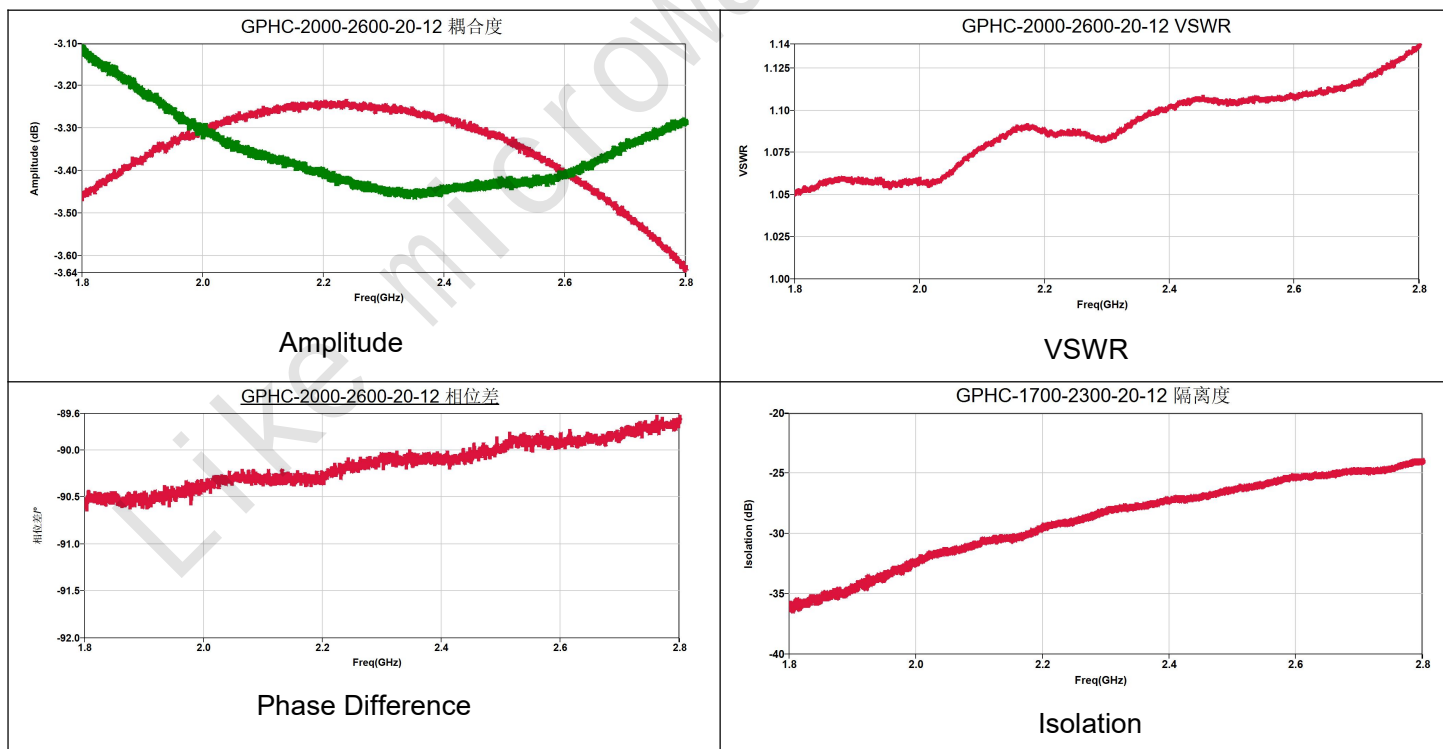


2.Typical Specifications

Function	Terminal	Function	Terminal
Sum Port	1	Isolation	3
Port 1(0°)	4	Port 2(90°)	6

Parameter	Frequency (MHz)	SPEC		
		Min.	TYP	Max.
Insertion Loss (dB)	2000~2600		0.6	0.9
VSWR			1.4	1.8
Amplitude Unbalance(dB)			0.8	1.2
Phase Unbalance (Degrees)			3	5
Isolation			20	15

3.Electrical Performance Specifications

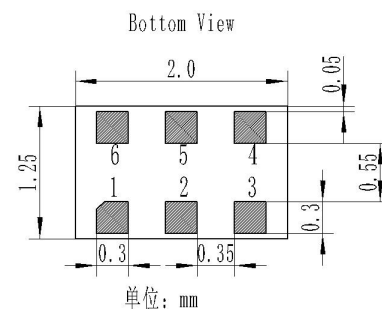


GPHC-3000-4000-20-12

PCB 3dB 90° Hybrid Coupler

1.Features

- 0805 (2.0×1.25mm) Stand Package
- Continuous wave 2W
- Operating temperature: -55℃ to +125℃

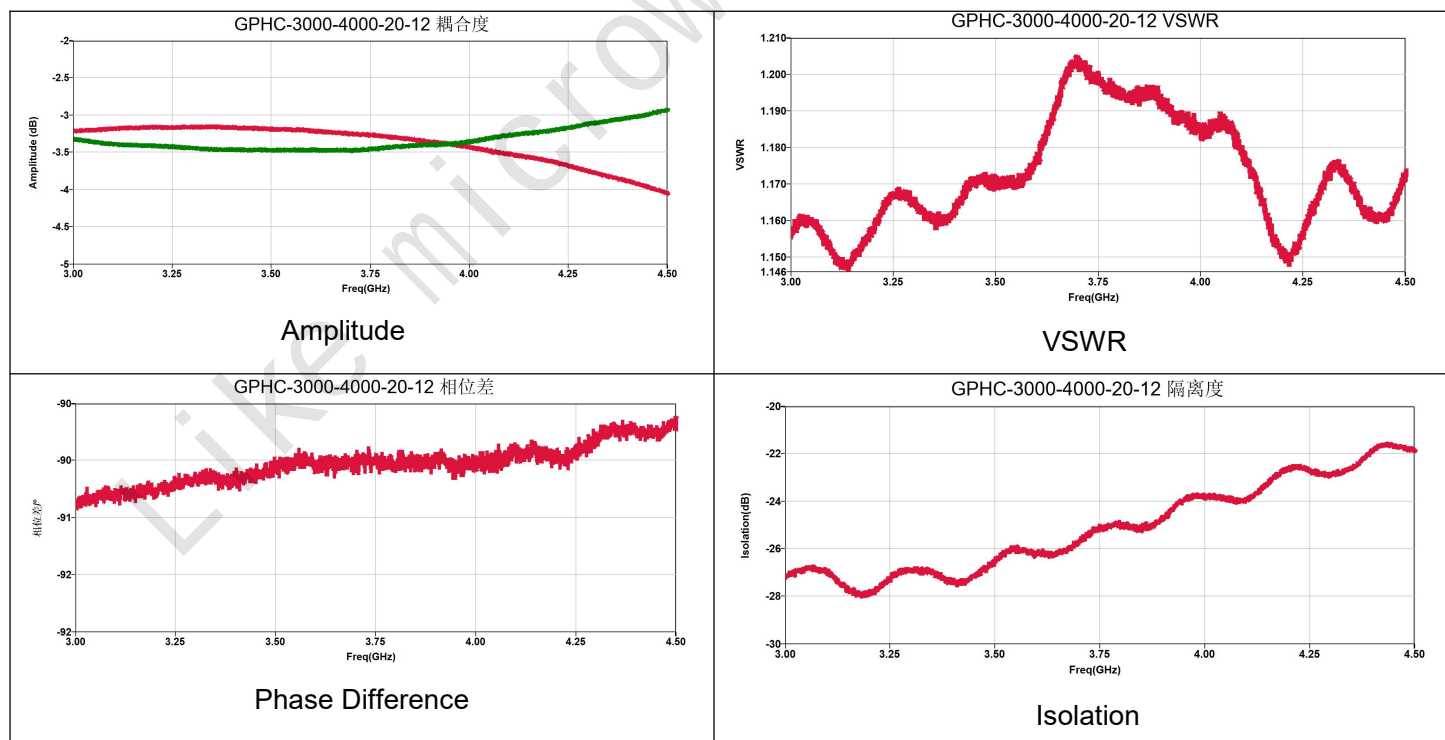


Function	Terminal	Function	Terminal
Sum Port	1	Isolation	3
Port 1(0°)	4	Port 2(90°)	6

2.Typical Specifications

Parameter	Frequency (MHz)	SPEC		
		Min.	TYP	Max.
Insertion Loss (dB)	3000~4000		0.6	0.9
VSWR			1.4	1.8
Amplitude Unbalance(dB)			0.8	1.2
Phase Unbalance (Degrees)			3	5
Isolation			20	15

3.Electrical Performance Specifications

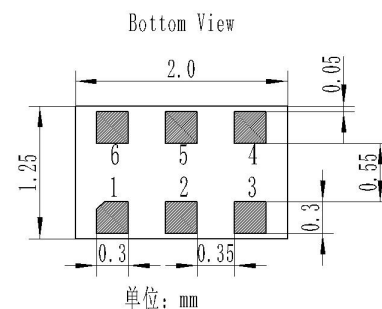


GPHC-3500-6000-20-12

PCB 3dB 90° Hybrid Coupler

1.Features

- 0805 (2.0×1.25mm) Stand Package
- Continuous wave 2W
- Operating temperature: -55℃ to +125℃

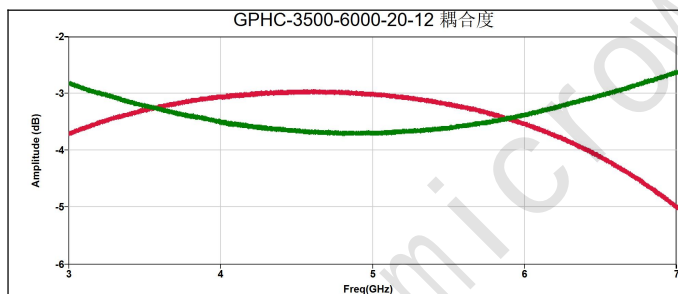


Function	Terminal	Function	Terminal
Sum Port	1	Isolation	3
Port 1(0°)	4	Port 2(90°)	6

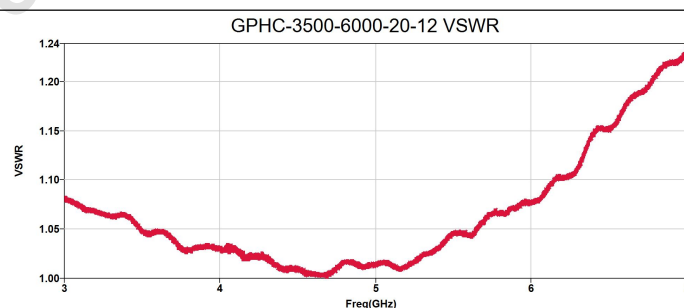
2.Typical Specifications

Parameter	Frequency (MHz)	SPEC		
		Min.	TYP	Max.
Insertion Loss (dB)	3500~6000		0.6	0.9
VSWR			1.4	1.8
Amplitude Unbalance(dB)			0.8	1.2
Phase Unbalance (Degrees)			3	5
Isolation			20	15

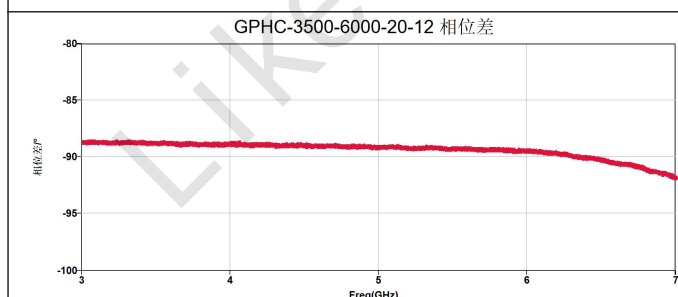
3.Electrical Performance Specifications



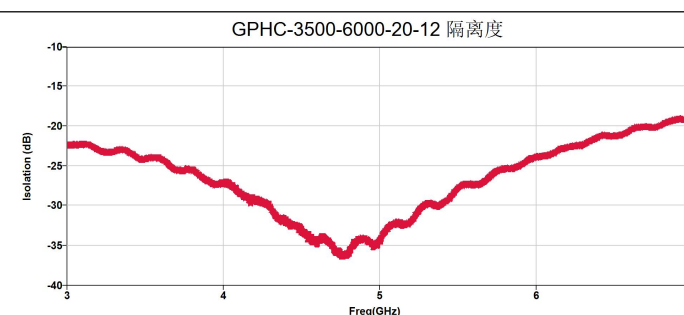
Amplitude



VSWR



Phase Difference



Isolation